

Event-Conditioned Crypto–Macro Spillover Networks

John Beecher

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Project Proposal

Problem Statement

Crypto markets increasingly trade as part of a broader macro-financial system. During ordinary periods, assets like BTC and ETH are primarily driven by crypto-native flows and sector-specific narratives. However, around scheduled macroeconomic information events (e.g., CPI and NFP releases, FOMC meetings), crypto tends to behave as a high-beta macro asset, reacting to shocks in rates, equities, and volatility. This project investigates whether the underlying dependency structure between crypto markets and traditional finance shifts during these windows. I aim to model this as a non-stationary network problem, identifying whether macro variables become “parents” of crypto variables during event windows.

A secondary goal is to test whether these event-conditioned dependency structures are useful for portfolio risk management. If crypto becomes more tightly connected to macro risk factors during scheduled information events, then a static allocation may underestimate concentration risk. I will therefore study whether event-conditioned network estimates can inform a more adaptive allocation rule, particularly through Hierarchical Risk Parity.

Relation to Course Material

This project utilizes the framework of Dynamic Bayesian Networks (DBNs) and Structure Learning. I will treat the market as a template model where edges represent both intra-slice (contemporaneous) and inter-slice (temporal) dependencies. The project focuses on learning the structure of a Gauss-Bayesian Network under non-stationary conditions, specifically utilizing score-based methods to identify conditional independence relationships.

The portfolio-risk extension also connects to graphical modeling because the learned dependency structure directly informs how assets should be grouped for risk allocation. If the event-conditioned DBN suggests that crypto variables become more dependent on macro variables during CPI or FOMC windows, then the corresponding portfolio should treat crypto exposure as part of a broader macro-risk cluster rather than as an independent diversifying asset.

Proposed Approach

I will implement a robust structure learning approach using the DYNOTEARS algorithm. This allows for the simultaneous estimation of lagged temporal edges and contemporaneous directed acyclic graphs (DAGs) by minimizing a penalized loss function subject to an acyclicity constraint.

- **Robust Estimation:** To handle the heavy-tailed nature of crypto returns and macro shocks, I will replace the standard squared loss with a Huber loss function. This provides robustness against outliers while maintaining a differentiable objective for the DYNOTEARS optimization.
- **Event Conditioning:** I will use fixed and observable windows around CPI, NFP, and FOMC releases to compare “Event” vs. “Ordinary” topologies.
- **Sliding Window Analysis:** I will implement a sliding window approach within event periods to track the evolution and decay of structural edges as macro information is digested by the market.
- **Sparsity:** I will use L_1 regularization and the Bayesian Information Criterion (BIC) to ensure the resulting spillover networks are sparse and interpretable.

As a portfolio-risk extension, I will use Hierarchical Risk Parity (HRP) to translate the learned dependency structure into an allocation rule. HRP is useful here because it does not rely heavily on expected-return estimates or direct covariance-matrix inversion. Instead, it uses the correlation structure among assets to form hierarchical clusters and then allocates capital recursively across those clusters according to their risk.

Given an estimated correlation matrix ρ , I will construct a distance matrix

$$d_{ij} = \sqrt{\frac{1 - \rho_{ij}}{2}},$$

then apply hierarchical clustering to identify groups of assets that behave similarly. I will compare the resulting clusters during ordinary periods and event windows. For example, if BTC and ETH cluster primarily with crypto-native variables during normal periods but move closer to Nasdaq, Treasury yields, or VIX during macro-event windows, this would provide evidence that crypto becomes more macro-sensitive during those periods.

I will then construct and compare several portfolios:

- **Static HRP:** A baseline HRP allocation estimated using the full sample.
- **Event-Conditioned HRP:** Separate HRP allocations estimated using ordinary, pre-event, and post-event windows.
- **Spillover-Informed HRP:** A dynamic rule that switches toward the event-conditioned HRP allocation when the learned DBN indicates high macro-to-crypto spillover risk.

This allows the project to connect the probabilistic graphical model to a practical trading and risk-management question: whether learned event-conditioned dependencies can improve portfolio allocation during macro-sensitive periods.

Datasets and Software

The dataset consists of time series for BTC, ETH, and SOL, alongside crypto-native derivatives data such as funding rates and perp-spot basis via exchange APIs including Binance and Hyperliquid. Macro variables include Treasury yield changes, CME 10Y futures, Nasdaq-100 proxies, and the VIX via Polygon.io and alpaca.markets.

For the HRP extension, the tradable asset universe may include BTC, ETH, SOL, Nasdaq or S&P 500 proxies, Treasury futures or ETFs, dollar proxies, and volatility proxies. Non-tradable

variables such as funding rates, perp basis, event indicators, or inferred spillover-risk scores will be used as conditioning information rather than directly receiving portfolio weights.

- **Software:** Primary modeling will use Python, including numpy, scikit-learn, scipy, networkx, and matplotlib.
- **Structure Learning:** I will use DYNOTEARS-style optimization for dynamic graph estimation and networkx for graph visualization.
- **Portfolio Construction:** I will implement HRP using scipy’s hierarchical clustering tools and custom portfolio-weighting code.
- **Data Engineering:** I will use Python, pandas, and polars for efficient ingestion and timezone alignment from UTC to ET, ensuring precise synchronization between continuous crypto trading and traditional market hours. If Python proves to be too slow for the amount of data we need to process, then I will consider using Rust as well.

Evaluation Metrics

- **Structural Hamming Distance (SHD):** I will use SHD to quantify the topological shift between ordinary-regime and event-regime graphs.
- **Stability Selection:** I will evaluate the persistence of edges across randomized data sub-samples to ensure structural results are not artifacts of specific outliers.
- **Predictive Likelihood:** Out-of-sample log-likelihood will be used to compare the event-conditioned DBN against a static baseline.
- **Clustering Stability:** For the HRP extension, I will compare whether asset clusters change meaningfully between ordinary periods and macro-event windows.
- **Portfolio Risk Metrics:** I will evaluate static HRP, event-conditioned HRP, equal-weight, and possibly standard risk-parity portfolios using realized volatility, maximum drawdown, Sharpe ratio, event-window loss, and turnover.

Anticipated Timeline

- **Week 1:** Data collection, alignment, and event windowing.
- **Week 2:** Implementation of DYNOTEARS with Huber loss; baseline graph estimation.
- **Week 3:** Sliding window structural analysis and SHD comparison across regimes.
- **Week 4:** HRP portfolio-risk evaluation, final report writing, and presentation prep.